

SKINNER'S SCIENTIFIC SPECTRUM

“An electron beam gun [...] would most definitely ionise your coffee.

- NEO SKINNER

MATHEMATICS

ROME VERSUS THE VATICAN

As December approaches, those of you particularly inclined to the giving season may already be searching through shops for the perfect festive gift. Looking past socks for the moment, another classic gift is the calendar, and with a plethora of different designs available, it's a wonder that I do not manufacture a novelty calendar of my own – sadly all calendars of that sort will not be endorsed by me. However, in 1752, purchasers of almanacks – the calendar-based font of knowledge of the time – across Britain opened to the September pages to find the following notice:

According to an Act of Parliament passed in the 24th Year of his Majesty's Reign, and in the Year of our Lord 1751, the Old Style ceases here, and the New takes place; and consequently the next Day, which in the Old Account would have been called the 3d, is now to be called the 14th; the 11 intermediate nominal Days from the 2d to the 14th being omitted, or rather annihilated, this Year; and the Month contains no more than 19 Days, as the Title at the Head expresses.

(Var., 1752, p.24) Full Text See p.10

But, what did this mean? It meant that Britain had gone Gregorian. In 1752, the *Calendar (New Style) Act 1750* replaced the Julian calendar with the Gregorian, stripping September of 11 days (Calendar (New Style) Act 1750).

It was Pope Gregory XIII who first introduced the Gregorian calendar in 1582 as an improvement over the existing calendar named in honour of Julius Caesar (Julian). The Julian calendar had a year fixed at 365.25 days (365 days and 6 hours), whilst the Gregorian calendar changed this to 365.2425 days (365 days, 5 hours, 49 minutes, and 12 seconds) by switching 100 leap years every 400 years for 97 years. The 365.2425 figure corrected the length of a calendar year against the length of a solar (tropical) year, bringing our calendar more in line with the physical world over long periods. In fact, the Julian calendar drifted by roughly 11 minutes every year and was out by 11 days in 1752, when Britain made the switch – troubling (Bikos and Kher, 2019; Jameson, 2022)!

Now, with the mathematical explanation out of the way, why is it important that our calendars be accurate? In the 16th century, events such as equinoxes and solstices were of great importance as they defined seasons and, with most of the populace farming, it was vital that seasons were well established for crop rotations and harvests.

Farming is actually the reason that ancient civilisations first developed calendars. Starting with the Aztecs, astronomy was used to map the movements of the stars, Moon, and Sun which they believed helped their crops to grow. Their theory for this may have been way off, but they hit onto the right area and their success with farming led them to thrive as a civilisation.

The Aztec calendar mapped the Sun, allowing them to identify different parts of the Earth's cycle (seasons) so that they could plant and harvest crops at the right times. But, it was not just planting and harvesting, the calendar further allowed the Aztecs to carefully control their irrigation systems to optimise their crop yield. Without the calendar, it seems as though modern civilisation(s) would not have emerged.



Pop Your Birthdays on There, an Aztec Calendar Stone

In 753 BCE, Romulus – a semi-divine child raised by a she-wolf, I cannot make this up if I tried – founded Rome (naming it after himself) and with it, introduced the first Roman calendar. The first Roman calendar built upon the calendar ideas of the ancient civilisations that had preceded it and contained just 10 months – 304 days – based on the cycle of the Moon, a *minor* oversight which led it to stray considerably from the solar year (History Skills, 2024b).

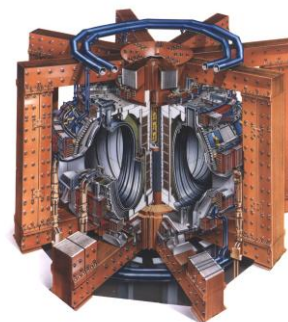
This oversight was ‘fixed’ around (c.) 700 BCE by Numa Pompilius, who added two months to the start of the calendar – points for those who guessed January and February – to make the year up to 355 days. This was still not quite enough for Julius Caesar who reformed the calendar yet again in 46 BCE which was further improved by Emperor Augustus around 8 BCE to ensure the months were consistent – their names are still embedded in our calendar to this day (History Skills, 2024b).

The calendar remained that way (in Britain) until 1752 when Parliament finally adopted the Gregorian calendar – 170 years after it had first been introduced.

PHYSICS

WHEN TWO BECOME ONE

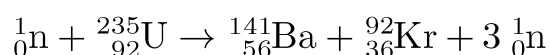
In an inconspicuous building just outside Oxford, physicists anxiously clamber to see the pink glow from the television sets in the control room. A cheer echoes around the room as other physicists eye the energy monitors, searching for glimpses of future hope. They had done it. The Joint European Torus (JET) was the first project to produce commercially viable controlled nuclear fusion power.



Red Reactor at Night, Shepherds Delight; LTR: Plasma in JET, JET Design

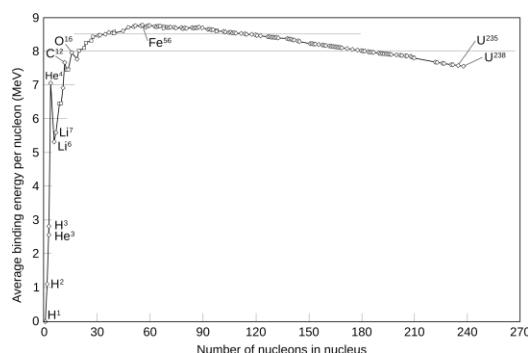
The JET project is just one in a long line of nuclear fusion projects aimed at researching the suitability of commercial nuclear power. But, before we dive into the commercial prospects, let us first look at what nuclear fusion is and what makes it different from its dangerous brother nuclear fission.

In nuclear fission, a heavy atom is bombarded with neutrons until it splits into two (or more) lighter atoms, for example, the fission of uranium-235 to form barium-141, krypton-92 and three neutrons:



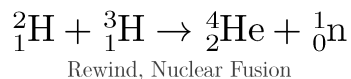
Splitting Hairs, Nuclear Fission

The three neutrons formed here are key as they allow nuclear reactors to achieve a chain reaction with neutrons released going on to split other uranium-235 atoms. The key part of this process is surrounding the nuclear binding energy – the energy holding the nucleus together [see issue 2] – which is lower for uranium than for the fission products, allowing energy to be released that is used as power.



Bound to Excite, the Binding Energy Curve

In nuclear fusion, we play back the process in reverse, with two light atoms combining to form a heavier one. In the proposed commercial nuclear fusion reactors, deuterium (hydrogen-2) and tritium (hydrogen-3) are forced together under high energy to form helium-4 and one extra neutron:



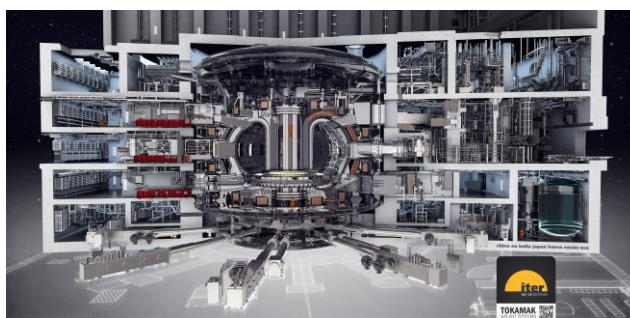
This process produces energy in the same way as nuclear fission, helium-4 has a higher binding energy than the two reactants and so the leftover energy can be harnessed for power. If you look at the binding energy curve above, you can see that we are approaching the maximum binding energy (turning point) from the other direction, so the exact same net result is reached.

However, there is one big problem: the high energy. In order to achieve nuclear fusion, two atoms need to be pushed together to overcome the Coulomb barrier – the limit of the repulsive electromagnetic force – so that the strong nuclear force pulls the two nuclei together to form a single atom product (plus a neutron). This is what requires a lot of energy and, here on Earth, the method to achieve this is very similar to that of stars.



Keep Your Ion It; LTR: A Jacob's Ladder, the Sun

The force of gravity within stars superheats the core and forms a plasma – the fourth state of matter consisting of delocalised ions – which has the energy needed to burst through the Coulomb barrier and fuse atoms. Fusion reactor designs employ this strategy to accomplish fusion and harness the energy released in the neutrons produced for power.



Big Plans, the ITER Project

The horizon is bright for nuclear fusion, with the next step already under construction in southern France. The ITER project promises to be the largest fusion reactor in the world and generate ten times its original input power – 500 MW from 50 MW (ITER Organisation, 2023).

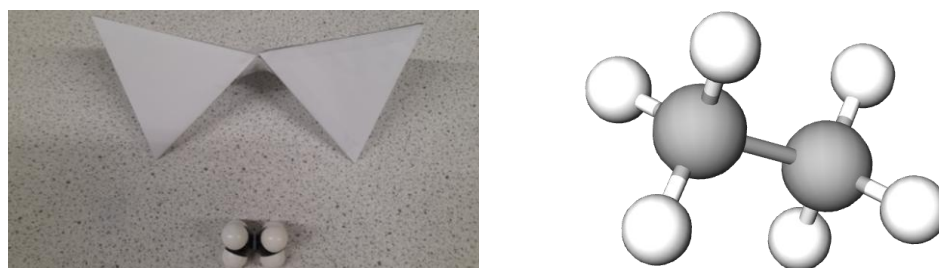
CHEMISTRY

STRIKE A POSE!

At chemistry's hottest fashion show, ethane struts down the runway – eyes turning at the height of fashion – the ball and stick outfit sure is an icon. But! Here comes ethene with a slimming wireframe, a modern outfit for a strict molecule. Different chemical models aid our understanding of complex intramolecular interactions – but what did we do before the digital era?

The fancy graphics that I use with moderate zeal within articles are a remarkably new outlook on molecular modelling; with a finished image file produced in a matter of minutes, the challenge of pre-digital chemistry is not something I would wish upon any chemistry author. But curiosity got the better of me and I dared myself to write an article using models only produced with paper – I succeeded, but added a few other models just for good measure...

The first challenge is what atoms look like using paper – what distinguishes carbon from oxygen or nitrogen from hydrogen? I decided to represent each atom with a shape so that each vertex was a valence (bonding) electron. Carbon is a tetrahedron (triangular-based pyramid) with four vertices for four valence electrons and, in the same way that every vertex represents a hydrogen atom in a skeletal formula, a tetrahedron alone designates methane (CH_4). It is only when the vertices of two shapes touch that hydrogen atoms vanish. The simplest example of this using only tetrahedrons is the lowly ethane (C_2H_6), with only six instead of the usual eight hydrogen atoms resulting from the two touching vertices.



The Thrills of Sticky-Back Plastic; LTR: Paper and *molymod* Ethane, Digital Ethane

With the fairly simple basics covered, it is easy to see how ethene (C_2H_4) might be constructed – just slap together two tetrahedrons at four vertices (or two edges)!



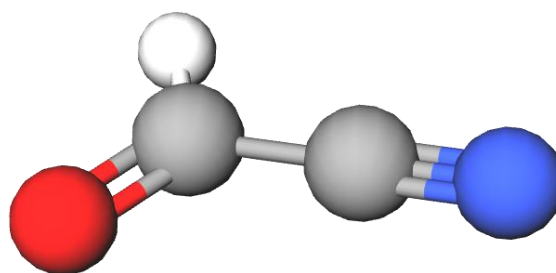
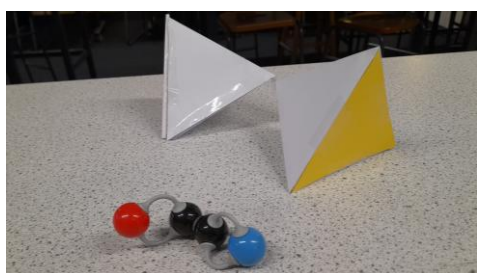
Spot the Difference; LTR: Paper and *molymod* Ethene, Digital Ethene

Using this framework, we can work out what represents nitrogen and oxygen in paper. Nitrogen has three valence electrons and is thus a triangle with three vertices whilst oxygen has two

bonding electrons forming a shape with two vertices. A shape with two vertices? Yes. For the purposes of this building exercise, a shape with two vertices is a paper straw – the two ends being the valence electrons of oxygen.

It is worth dwelling here on the two bonding electrons of oxygen. You see, whilst oxygen actually has six electrons in its outermost electron shell, only two of these participate in covalent bonding. This is because the rule of thumb for the bonding of low p-block elements is the octet rule which states that atoms will bond to achieve eight electrons in their outermost shell. Oxygen has four electrons that are unreactive and two that readily react to each form one covalent bond with a shared electron pair. The simplest example of this is water (H_2O) where hydrogen contributes one electron to each of the covalent bonds formed by the two bonding electrons in oxygen – this is a boring example using a single straw which you may very well already have in the kitchen cupboard of your home.

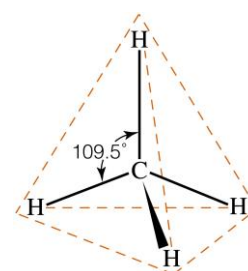
A slightly more interesting example is methanoyl cyanide (C_2HNO , preferred name: formyl cyanide) with an oxygen and a nitrogen bound to either end of an ethane molecule to form the simplest acyl cyanide, exciting!



Modern Architecture at its Finest; LTR: Paper and *molymod* Formyl Cyanide, Digital Formyl Cyanide

With this basis, almost all organic compounds can be made from paper – their four building blocks being carbon, hydrogen, nitrogen, and oxygen. By a simple extension to this system, we can derive shapes for all s- and p-block elements, although it is worth noting that this system only really works for covalent compounds. Ionic compounds require a modified system where touching vertices represent electron transfer and metallic compounds are an entirely different kettle of fish due to the delocalised electrons.

Throughout this article, I have shown digital and physical (*molymod*) ball and stick models alongside my paper creations to show how the molecules I have discussed are normally modelled. With a bit of imagination, the ball and stick models can be found within the paper models as demonstrated by methane (right).



But, these are far from the only ways of modelling molecules, and I have to, unfortunately, leave you on a cliffhanger until next week as I have run out of space in this issue! Join us again next month to examine the world of quantum molecular modelling – this could get rather messy...

BIOLOGY

ALGAE BONANZA!

Have you ever seen a pond or body of water that looked like a thick, green soup? That was probably an algal bloom. This occurs when conditions are just right for a baby boom! In marine environments, these tend to be caused by dinoflagellates, part of the Protozoa phylum, like *Karenia brevis*, responsible for the ‘red tides’ sometimes seen in summer. In freshwater environments, cyanobacteria dominate; these are photosynthetic bacteria rather than algae, but come under the algal bloom ‘umbrella’. Periodic rapid expansion of algal populations is a key event in ocean ecosystems, providing essential nourishment for krill and higher trophic levels. However, when these ‘blooms’ have the potential to wreak havoc on ecosystems or interfere with human activity, they are called ‘Harmful Algal Blooms’, HABs.



Your Beach Day Out, Spoilt by Algae's Own Party; LTR: Algae in Boracay, a Reverse Red Sunset

HABs occur due to an interplay of natural and human causes. Natural phenomena like abnormally slow water circulation, extreme weather events (hurricanes, floods, drought), or unusually high water temperatures can lead to an algal bloom. More and more, humans are increasing the likelihood of HABs by pumping nutrients into waterways via agricultural runoff or wastewater effluent. As temperatures spike, especially during late summer, these conditions create a perfect storm for algae to thrive. Climate change will only increase HABs' frequency, distribution, and magnitude.

To outline but a few expected impacts, increasing extremes in rainfall could lead to more frequent peaks in surface runoff and therefore, sudden nutrient loading into oceans and lakes. Surges in nutrient levels often cause eutrophication, as algae (toxic or not) rapidly reproduce. Once water carbon dioxide (CO₂) concentrations are depleted, the algae die off and decompose, creating a hypoxic environment (lacking oxygen). Alongside this, the algae can clog the gills of fish and invertebrates, cutting off vital oxygen, or smother corals and underwater vegetation. These effects can result in large-scale die-off of aquatic organisms, impacting commerce, food security, and the ecosystem's natural balance. In landlocked waters, subsequent droughts could intensify this by sustaining high nutrient levels and fuel more resilient, often toxin-producing algae.

Similar effects to eutrophication can be seen in coastal areas after extreme rainfall, where the ‘lighter’ freshwater sits on top of denser saltwater. This naturally occurs when fast-flowing estuaries discharge into an area with weak tidal currents, known as a ‘saltwater wedge.’ But

suppose a wedge forms suddenly, the oxygen-rich freshwater will suffocate algae in the water's upper layers, which sink and decompose in the saline, deeper waters, creating another hypoxic environment.



Getting Your Salty Foot in the Door; LTR: Algal Bloom in Lake Erie (Ohio, USA, 2011), a Salt Wedge Estuary

Climate change could also increase the competitive fitness of toxic HAB-causing algae compared to other non-toxic species. For example, toxic HAB-causing cyanobacteria or 'cyanoHABs' are adapted to still, warm waters, moving vertically through the water column better than other species to collect nutrients from cooler, dark layers of water, for photosynthesis in warmer, brightly lit surface waters. Rising seas will create large areas of shallow, stable coastal waters that suit cyanoHABs. Also, acidification caused by increased air and water carbon dioxide concentrations will favour cyanoHABs and reduce grazing by consumers. That is to say, climate change means HAB-causing algae will be able to grow much faster than other species can compete with.

Intensifying drought and increased evapotranspiration have and will continue to drive up the salinity of inland waters. Combined with saltwater invasion with higher sea levels and changes in coastal upwelling (where offshore winds cause deep, nutrient-rich waters to rise to the surface), marine algae are starting to contaminate freshwater supplies. Both problematic for human consumption and the environment, these blooms in vulnerable 'naïve' ecosystems will be particularly damaging, as the community has never had this selective pressure to develop resistance to the algae. This matters because some algae produce toxins that can directly kill fish and mammals and cause serious illness or even death in humans. These toxic algae also cause shellfish poisoning, leading to severe gastrointestinal and neurological symptoms in humans. This poses not only serious health risks but commercial too, as fish and shellfish stocks may decline, and what is caught could be unsellable.

Swamped by news stories focussing on the novel effects of climate change, it is easy to overlook what has happened for decades. Since the early 2000s, saltwater invasion of freshwater lakes in the US has seeded 'golden algae' blooms and devastated freshwater fish populations (EPA, 2024a). While we can deploy measures to mitigate the worst impacts, these blooms are but another symptom of larger issues: climate change and excessive chemical fertiliser use. As a society, we must act in our daily lives and push industries and governments to lead the fight against the destruction of our natural world.

RECOMMENDATIONS

A-LEVEL WIDER INDEPENDENT STUDY

- 📖 **Mathematics:** The History of Logarithms.
- 📖 **Physics:** Are Tachyons Really Possible?
- 📖 **Chemistry:** Napoleon and Scheele's Green.
- 📖 **Biology:** Microplastics As Endocrine Disruptors

DISCRETE FURTHER READING

- 📖 **Mathematics:** New Scientist October 2024, Amateur sleuth finds largest known prime number with 41 million digits [↗](#)
- 📖 **Physics:** ScienceAlert October 2024, We Finally Know Why Ancient Roman Concrete Was Able to Last Thousands of Years [↗](#)
- 📖 **Chemistry:** Scientific American October 2024, Challenging Big Oil's Big Lie about Plastic Recycling [↗](#)
- 📖 **Biology:** Nature October 2024, How does overeating lead to diabetes? A surge of neurotransmitters [↗](#)

CONTINUOUS FURTHER STUDY TOPICS

- 📖 **Mathematics:** Calculating Lunisolar Calendars
- 📖 **Physics:** How Do We Confine Plasma?
- 📖 **Chemistry:** What Reacts With Acyl Cyanides?
- 📖 **Biology:** Formation of the Intracellular Cytoskeleton

BOOK APPRAISAL

Stewart, I. (2012). *Seventeen Equations That Changed the World*. London: Profile.

EXTRA – NOBEL PRIZE CROSSWORD

Chemistry World October 2024, Nobel prize crossword 2024 [↗](#)

MONTHLY PROBLEM

Solve for x :

$$\frac{x(x(x(x(x-20)+160)-640)+1280)-1024}{(x-4)^4} = 5$$

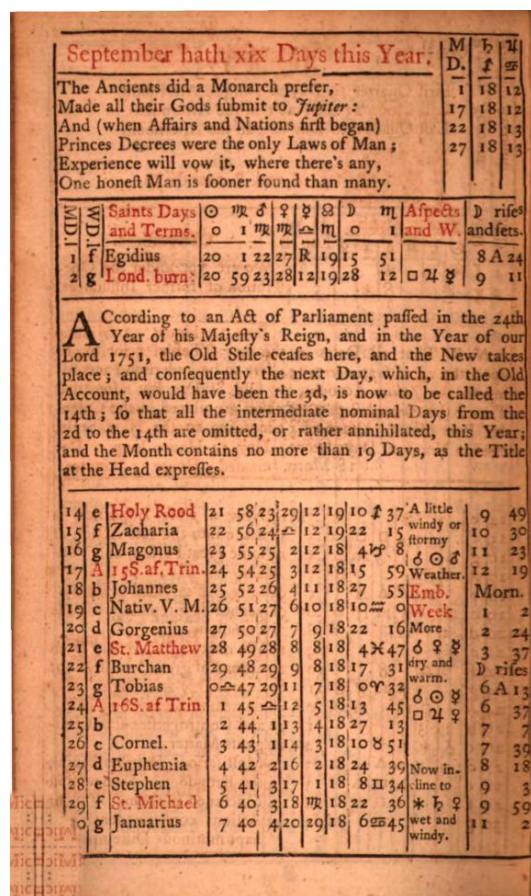
Solution next month.

Last month's solution:

$$x = \frac{1}{2}$$

CREDITS

ADDITIONAL MATERIAL



The 1752 almanack by J. Goldsmith featured the extract used within the mathematics article; a full version of the almanack can be found online on Google Books, a link is provided in the sources and provenance section below.

Month (Latin)	Days
Martius	31
Aprilis	30
Maius	31
Iunius	30
Quintilis	31
Sextilis	30
September	30
October	31
November	30
December	30

The original Romulan Roman calendar featured 10 months of either 30 or 31 days, leading to a total year length of 304 days – although it may have been that winter was assigned extra days at the end of the year. Needless to say, this

needed significant reform – multiple times in fact! (Jasiński, 2024)

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- ☞ Webb, P. (2024). *Figure 13.6.7 A salt wedge estuary. Strong river outflow (light blue arrows) creates a layer of mostly fresh water that sits on top of a wedge of encroaching seawater along the bottom (dark blue arrow). Salinities are in ppt (JR)*. [digital] *Geosciences LibreTexts*. Available at:

[geo.libretexts.org/Bookshelves/Oceanography/Introduction to Oceanography \(Webb\)/13%3A Coastal Oceanography/13.06%3A Estuaries](https://geo.libretexts.org/Bookshelves/Oceanography/Introduction_to_Oceanography_(Webb)/13%3A_Coastal_Oceanography/13.06%3A_Estuaries) .

- ☞ Wiessinger, S. and NASA's Scientific Visualization Studio (2024). *Sun Releases Two Strong Flares on February 21 and 22 (SVS14533 - 20240221 230634 4096 0171)*. [digital] *Wikimedia Commons*. Available at: [commons.wikimedia.org/wiki/File:Sun Releases Two Strong Flares on February 21 and 22 \(SVS14533 - 20240221 230634 4096 0171\).jpg](https://commons.wikimedia.org/wiki/File:Sun_Releases_Two_Strong_Flares_on_February_21_and_22_(SVS14533_-_20240221_230634_4096_0171).jpg) .

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- ☞ Equations modified for publishing by *Lagrida*

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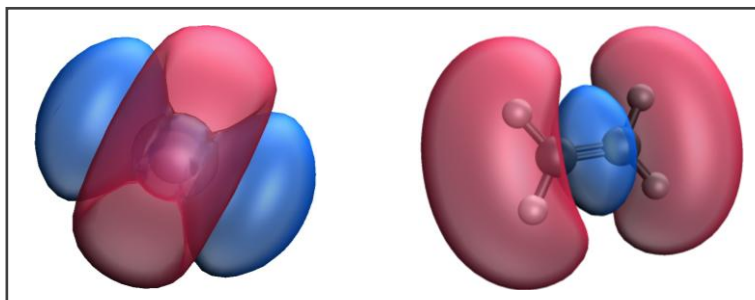
The editor extends their gratitude to Spiring Enterprises Limited, the makers of the *molymod* molecular modelling system, for their help with the chemistry article of this issue.

CORRECTIONS

- ☞ **Issue 5 – Mathematics:** Mr. S. Beakes identified that one of the in-text equations should read “ $4 \times 3 = 12$ ” instead of “ $4 + 3 = 12$.” Great work!

The editor can be contacted regarding errors and suggestions via the email skinners-science-spectrum@magna-aspirations.org .

Looks like I'll be continuing with chemical modelling next month...



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